

Student Checklist (1A)

This form is required for ALL projects.

1. a. Student/Team Leader: Amaar Chughtai Grade: 11
Email: amaar2cool@gmail.com Phone: 4089306097
b. Team Member: _____ c. Team Member: _____
2. Title of Project: Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's
3. School: Valley Christian High School School Phone: 4085132400
(if multiple schools, list of the team leader or list all schools).
School Address: 100 Skyway Dr #130, San Jose, CA 95111
4. Adult Sponsor: Mr. John Crandall Phone/Email: jcrandall@vcs.net
5. Does this project need SRC/IRB/IACUC or other pre-approval? ☐ Yes ☒ No Tentative start date: 01/12/26
6. Is this a continuation/progression from a previous year? ☐ Yes ☒ No
a. If yes, attach the previous year's ☐ Abstract **and** ☐ Research Plan/Project Summary
b. Explain how this project is new and different from previous years on
☐ Continuation/Research Progression Form (7); include forms for all previous years
7. This year's experimentation/data collection (include forms for all previous years):
01/12/26 02/15/26
Actual Start Date: (mm/dd/yy) End Date: (mm/dd/yy)
8. Where will you conduct your experimentation? (check all that apply)
☐ Research Institution ☐ School ☐ Field ☒ Home ☐ Other: _____
9. Source of Data:
☐ Collected self/mentor ☒ Other List all URL(s) in Research Plan: <https://openneuro.org/datasets/ds005048/versions/1.0.1>
10. List the name and address of all non-home and non-school work site(s), whether you worked there virtually or on-site:
Name _____
Address: _____
Phone/email _____
11. **Complete a Research Plan/Project Summary following the Research Plan/Project Summary instructions and attach to this form.**
12. **An abstract is required for all projects after experimentation.**

Research Plan/Project Summary Instructions

A complete Research Plan/Project Summary is required for ALL projects and must accompany Student Checklist (1A).

1. The Research Plan is to be written prior to experimentation following the instructions below to detail the rationale, research question(s), methodology, and risk assessment of the proposed research.
2. If changes are made during the research prior to competing in an affiliated fair, such changes can be added to the original research plan as an addendum, recognizing that some changes may require returning to the IRB or SRC for appropriate review and approvals. If no additional approvals are required, this addendum serves as a project summary to explain research that was conducted.
3. If no changes are made from the original research plan, no project summary is required.
 - Some studies, such as an engineering design or mathematics projects, will be less detailed in the initial project plan and will change through the course of research. If such changes occur, a project summary that explains what was done is required and can be appended to the original research plan.
 - The Research Plan/Project Summary should include the following:
 - a. **RATIONALE:** Include a brief synopsis of the background that supports your research problem and explain why this research is important and if applicable, explain any societal impact of your research.
 - b. **RESEARCH QUESTION(S), HYPOTHESIS(ES), ENGINEERING GOAL(S), EXPECTED OUTCOMES:** How is this based on the rationale described above?
 - c. Describe the following in detail:
 - **List of materials:**
 - **Procedures:** Detail all procedures and experimental design including list of materials, methods for data collection, and when applicable, the source of data used. Describe your project delineating what you will do and what will be done by your mentor.
 - **Risk and Safety:** Identify any potential risks and safety precautions needed.
 - **Data Analysis:** Describe the procedures you will use to analyze the data/results.
 - d. **BIBLIOGRAPHY:** List major references (e.g. science journal articles, books, internet sites) from your literature review. If you plan to use vertebrate animals, one of these references must be an animal care reference.

Items 1–4 below are subject-specific guidelines for additional items to be included in your research plan/project summary as applicable.

1. Human participants research:

- a. **Participants:** Describe age range, gender, racial/ethnic composition of participants. Identify vulnerable populations (minors, pregnant women, prisoners, mentally disabled or economically disadvantaged).
- b. **Recruitment:** Where will you find your participants? How will they be invited to participate?
- c. **Methods:** What will participants be asked to do? Will you use any surveys, questionnaires or tests? If yes and not your own, how did you obtain? Did it require permissions? If so, explain. What is the frequency and length of time involved for each subject?
- d. **Risk Assessment:** What are the risks or potential discomforts (physical, psychological, time involved, social, legal, etc.) to participants? How will you minimize risks? List any benefits to society or participants.
- e. **Protection of Privacy:** Will identifiable information (e.g., names, telephone numbers, birth dates, email addresses) be collected? Will data be confidential/anonymous? If anonymous, describe how the data will be collected. If not anonymous, what procedures are in place for safeguarding confidentiality? Where will data be stored? Who will have access to the data? What will you do with the data after the study?
- f. **Informed Consent Process:** Describe how you will inform participants about the purpose of the study, what they will be asked to do, that their participation is voluntary and they have the right to stop at any time.

2. Vertebrate animal research:

- a. Discuss potential ALTERNATIVES to vertebrate animal use and present justification for use of vertebrates.
- b. Explain potential impact or contribution of this research.
- c. Detail all procedures to be used, including methods used to minimize potential discomfort, distress, pain and injury to the animals and detailed chemical concentrations and drug dosages.
- d. Detail animal numbers, species, strain, sex, age, source, etc., include justification of the numbers planned.
- e. Describe housing and oversight of daily care.
- f. Discuss disposition of the animals at the end of the study.

3. Potentially hazardous biological agents research:

- a. Give source of the organism and describe BSL assessment process and BSL determination.
- b. Detail safety precautions and discuss methods of disposal.

4. Hazardous chemicals, activities & devices:

- a. Describe Risk Assessment process, supervision, safety precautions and specific methods of disposal.
- b. Safety Data Sheets are not necessary to submit with paperwork.

Engineering Project Detailed Research Plan

Please complete the information/questions begun/seen below in red ink. Save your plan to your computer and upload a copy to your application Forms Folder.

Date: 01/08/26

Student Name: Amaar Chughtai

Project Title: Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's

Parts of the generic engineering project are listed below with descriptions to the students in the boxes. Students may provide a detailed research plan by describing their specific project in response to each box below.

Engineering Goal: PROBLEM BEING ADDRESSED: All engineering projects solve a problem or fill a need. This goal should be a simple statement that describes the product being designed, the customer it is for and the problem or need it satisfies. Example "The goal is to design a solar powered lawn mower for inexpensive automated lawn care for homeowners"

My Project Goal is:

Existing 40 Hz auditory entrainment mechanisms are typically delivered on a fixed schedule, where patients are given continuous auditory stimulation, and then predetermined breaks. The issue is Alzheimer Disease patients' responses to the stimuli weaken over time and differ across people. The goal of this project is to design and test a closed-loop software controller that uses live EEG feedback of an Alzheimer's patient listening to a 40 Hz auditory stimulus to decide whether the stimulus should stay on or pause for rest, such that it maximizes theta-gamma phase amplitude coupling for the patient.

Design Criteria: Design criteria define the product's required performance . Examples: " It will have a minimum speed of 10 KPH", The output will be within 15% of the mean of the experimental data". "It must withstand 15 repetitions of a 10N impact" The International System of units (SI) required.

My Project Design Criteria are the following:

The controller must output a stimulation decision (on or rest) every 1.0 second. The entire computation process including feature extraction and model inference for the predicted phase amplitude coupling score and the policy decision should complete within 0.5 seconds on a normal CPU. The EEG data should be processed in a continuous sliding window of 2.0 seconds. The model will predict the theta-gamma band phase-amplitude coupling (PAC) score computed using theta phase in 4-8 Hz and gamma amplitude in a fixed band around 40 Hz. Additionally, the decision of the controller should last for at least 5.0 seconds before changing. Most importantly, via simulation, the closed-loop schedule must outperform the fixed schedule by increasing mean PAC during stimulation by $\geq 15\%$.

Constraints: Constraints are factors that limit the engineer's flexibility such as size, cost, and time limitations. Examples: "It must fit in a box no larger than 10x20x50 cm" "The maximum cost is \$50" "The software must run in real time on a Raspberry Pi"

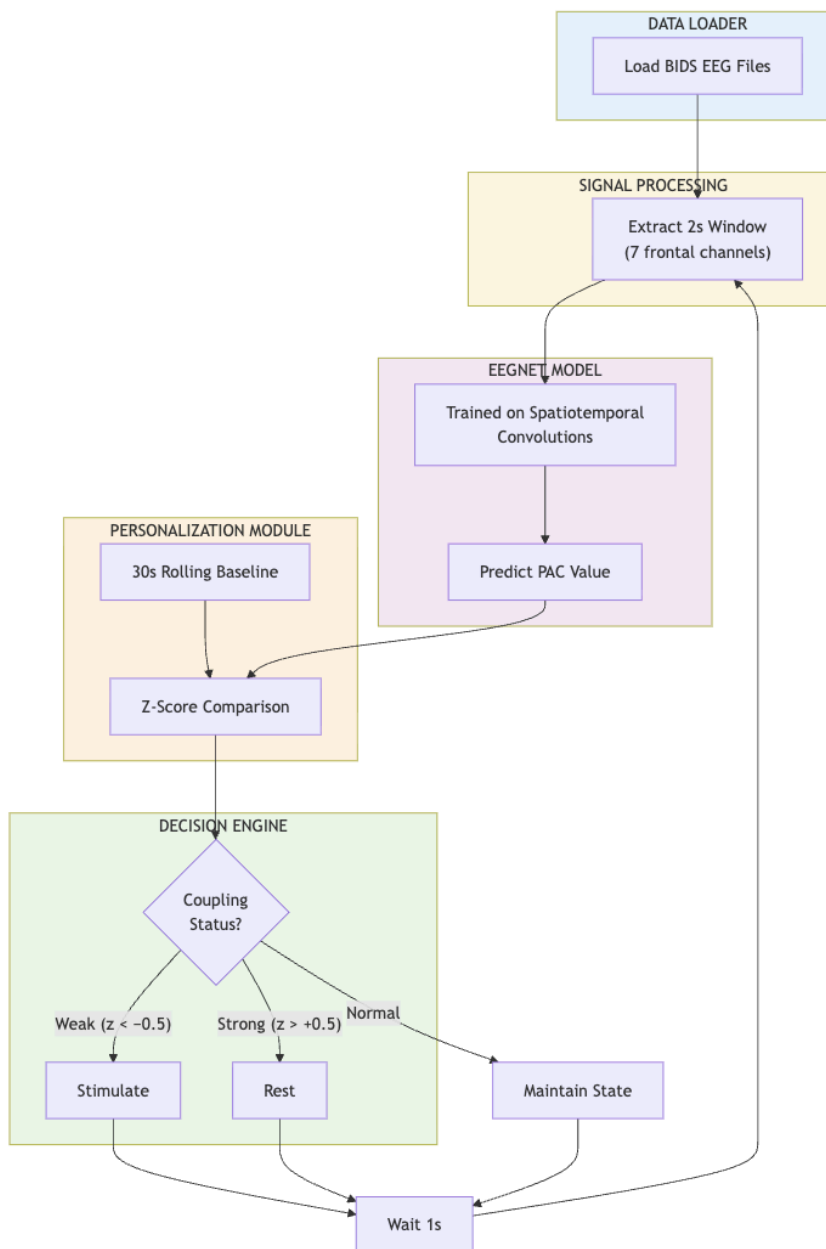
My Project Constraints are the following:

The project is limited to the public dataset OpenNeuro ds005048. The controller system is constrained to selecting only stimulation on or rest, and it cannot modify other stimulation parameters like frequency, amplitude, or waveform, the 40 Hz is fixed to the dataset protocol. The closed-loop model must carry out its prediction using only EEG samples up to the current time. Additionally, the closed-loop model must be trainable on my RTX 3080 GPU in reasonable time so I can test the model. The system must be able to run and compute in real time on a standard CPU in accordance with the time constraints in the design criteria specified above.

Provide your chosen design. For hardware, provide a sketch. For software, provide a flowchart. Indicate the components you will develop, and the libraries you are using.

My Project Design is shown below: insert photos, diagrams, or illustrations below.

The closed-loop controller consists of 5 main components. I will design a data loader to intake the BIDS EEG files, a signal processing unit for computing phase amplitude coupling labels, an EEGNet deep learning model for predicting PAC from 2 second windows of EEG, a personalization module implementing patient baseline adaptation, and a decision engine for stimulation control. I will be using PyTorch as the deep learning framework to build and train EEGNet. For loading the EEG, I will use MNE-Python. For the bandpass filtering for theta and gamma powers I will use SciPy, and NumPy will be used for almost all numerical operations. Pandas will handle the event files and allow me to visually display PAC accuracies. Together, the EEGNet model will receive 2.0 second windows from the 7 frontal electrode channels, predict PAC files, compare the prediction to the patient's 30 second baseline, and then output its stimulation decision when the coupling is beneath the patient's personalized threshold.



Test and evaluate your prototypes against the design criteria listed above to show how well the product meets the need/goal. Provide a test plan describing how you will test the design criteria and constraints you listed above., How will you analyze the data? If the product requires human testing please fill out and append <https://science-fair.org/wp/wp-content/uploads/2015/10/Research-Plan-Human-Participants.docx>

I test and analyze my prototypes using the following methods:

I will test the closed-loop controller by replaying pre-recorded EEG data from the OpenNeuro dataset. At each simulated decision point, the controller accesses only the data up to the time it is at, and outputs whether the auditory stimulus should be played or paused. To simulate how the brain responds to these decisions, I will extract the PAC response dynamics from the recorded 40 second stimulus periods and 20 second rest periods already in the dataset; when the controller decides to rest, I apply the observed rest response rate. I will run the simulation on 7 test subjects not used during EEGNet training to prevent overfitting. Most importantly, I will compare my controller's PAC optimization and total stimulation time against the scheduled stimulus' performance metrics that was used in the original study.

Bibliography: List at least five (5) major references (e.g. science journal articles, books, internet sites & dates of review) from your literature review. If you plan to use vertebrate animals, one of these references must be an animal care reference.

Example: Author's Name, Year of publication, "Quoted Title of Magazine Article (magazines only)"; Underlined *Title of Book or Magazine, date, volume, and number of magazine issue.* Page numbers read.

If you use a web site: www.urlname.ext, name of topic from the home page, author, and date read.

1. Shakya R, Suffczynski P, Shrestha S, Dangol S, Poudyal P, Mansur DI. 40 Hz binaural beats entrainment enhances the mood and cognition of medical students. *Int J Neurosci*. 2026 Jan;136(1):24-36. doi: 10.1080/00207454.2024.2429495. Epub 2024 Nov 19. PMID: 39535938. DOA 12/18/25
2. Martorell AJ, Paulson AL, Suk HJ, Abdurrob F, Drummond GT, Guan W, Young JZ, Kim DN, Kritskiy O, Barker SJ, Mangena V, Prince SM, Brown EN, Chung K, Boyden ES, Singer AC, Tsai LH. Multi-sensory Gamma Stimulation Ameliorates Alzheimer's-Associated Pathology and Improves Cognition. *Cell*. 2019 Apr 4;177(2):256-271.e22. doi: 10.1016/j.cell.2019.02.014. Epub 2019 Mar 14. PMID: 30879788; PMCID: PMC6774262. DOA 12/20/25
3. Lahijanian, M., Aghajan, H. & Vahabi, Z. Auditory gamma-band entrainment enhances default mode network connectivity in dementia patients. *Sci Rep* **14**, 13153 (2024). <https://doi.org/10.1038/s41598-024-63727-z> DOA 12/20/25
4. Tort AB, Komorowski R, Eichenbaum H, Kopell N. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *J Neurophysiol*. 2010 Aug;104(2):1195-210. doi: 10.1152/jn.00106.2010. Epub 2010 May 12. Erratum in: *J Neurophysiol*. 2010 Oct;104(4):2302. PMID: 20463205; PMCID: PMC2941206. DOA 12/21/25
5. Yang YH, Chang HW, Chien CF, Hour TC. Investigating the effects of 40 Hz sound stimulation on Alzheimer's disease pathways: Modulation of amyloid- β 42 secretion, tau phosphorylation, phagocytosis, and autophagy. *J Alzheimers Dis*. 2025 Jul;106(1):184-

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